Duration of antibody protection project

Nagoya-Oxford workshop 2026

August 2026

Abstract

How much longitudinal antibody data are needed before we can make a credible statement about the duration of vaccine protection? This workshop project will approach that question through simulation and model-based extrapolation. The working application is COVID-19 vaccination, for which both antibody-kinetics models and quantitative links between neutralisation and protection are available. The broader aim is to understand a recurring design problem: the long-term feature of a biological curve is often the feature least informed by a short study. The choices below are starting points for discussion, not a fixed analysis protocol.

1 Project question and aims

Consider a vaccine study that measures an antibody marker repeatedly after vaccination. Antibody levels often fall quickly at first and more slowly later. We would like to estimate when the marker will fall below a protection-relevant level, even when that crossing occurs beyond the observed follow-up. Immune correlates require careful terminology—a correlate can predict protection without being its sole cause—but they can still support useful quantitative predictions (Plotkin and Gilbert 2012; Earle et al. 2021).

The central question is therefore:

> *How long must participants be followed, and how informative must their longitudinal measurements be, before the duration of antibody-associated protection can be estimated credibly?*

The project has four linked aims:

1. describe plausible short- and long-term antibody trajectories;
2. quantify how much extrapolation is being asked of a fitted model;
3. understand when apparently similar fits imply very different durations of protection;
4. translate those findings into practical guidance for longitudinal study design.

COVID-19 provides a well-developed starting point: neutralisation is predictive of protection, and published work connects titres, variants, boosting and vaccine effectiveness (Khoury et al. 2021; Cromer et al. 2022; Gilbert et al. 2022; Feng et al. 2021). Whether the same framework transfers to other pathogens, assays and types of immune marker is deliberately left open for the workshop.

2 What do we mean by duration of protection?

Let $c_i(t)$ denote the latent antibody concentration for participant i at time t, and let c_{thr} be a protection-relevant threshold on the same assay scale. A natural individual-level target is the first threshold-crossing time

$$T_i^* = \inf\{t \geq 0 : c_i(t) \leq c_{\text{thr}}\}. \tag{1}$$

This quantity is a model-based estimand, not a directly observed biological expiry date. It can be undefined if the trajectory approaches a plateau above the threshold, and it depends on the endpoint and threshold definition.

For the initial COVID-19 application, one option is to anchor c_{thr} to the neutralisation level associated with 50% protection in the logistic relationship estimated by Khoury et al. (2021). The same fitted antibody curve could instead be passed through the full titre-to-protection relationship to obtain a protection curve $E_i(t)$. This makes explicit that "antibody duration" and "protection duration" are related but not interchangeable.

Heterogeneity creates a second distinction. The distribution of individual T_i^* values is not equivalent to the threshold-crossing time of a population-average antibody curve. A population target might instead be the time at which the fraction of people still above the threshold falls below a chosen level. The workshop should keep individual and population questions separate.

3 Candidate trajectory models

The candidate set is intended to expose different assumptions about the unobserved tail. No single family needs to be endorsed at the outset.

Single exponential

$$c(t) = c_0 e^{-kt}. \tag{2}$$

This is the simplest reference model. The log concentration has a constant slope and $T^* = k^{-1}\log(c_0/c_{\text{thr}})$ when $c_0 > c_{\text{thr}}$. It can be identified from relatively short follow-up if the trajectory is truly exponential, but it extrapolates the early slope indefinitely when that assumption is wrong.

Biphasic exponential

$$c(t) = Ae^{-k_1 t} + Be^{-k_2 t}, \qquad k_1 > k_2 > 0. \tag{3}$$

The fast component can represent contraction of short-lived antibody-secreting cells and the slow component a longer-lived population. This family is common in vaccine-antibody studies (M. T. White et al. 2015; Hogan et al. 2023). Its slow rate governs long-term duration but may be weakly identified while the fast component dominates. A smooth parameterisation in terms of peak level, fast and slow half-lives, and a switching time is convenient for simulation.

Power-law decay

$$c(t) = c_0(1 + \alpha t)^{-\beta}. \tag{4}$$

This curve decelerates continuously. It can resemble an exponential over a short interval while implying a much longer tail. Long-term antibody data provide biological motivation for considering decelerating decay rather than a constant half-life (Amanna et al. 2007).

Plateau or set-point

$$c(t) = A_\infty + (c_0 - A_\infty)e^{-kt}. \tag{5}$$

Here the long-term level is non-zero. A crossing exists only when $A_\infty < c_{\text{thr}}$. Short data may not distinguish a genuine plateau from a very slow decline, making "never crosses" an informative possibility rather than a numerical failure.

Mechanistic antibody-production model

A richer generating model can represent short- and long-lived antibody-secreting cells and antibody clearance separately:

$$\frac{dP_s}{dt} = -\delta_s P_s, \qquad \frac{dP_l}{dt} = -\delta_l P_l, \qquad \frac{dc}{dt} = \rho_s P_s + \rho_l P_l - \delta_c c. \tag{6}$$

When antibody clears quickly relative to the cells, this approaches biphasic decay; added memory or replenishment can generate an even slower tail. Related models have been developed for plasma cell dynamics and vaccine responses (Andraud et al. 2012; Pasin et al. 2019). Such a model may be most useful as a deliberately richer data-generating mechanism rather than as the default fitted model.

4 Observation and population parameterisation

Observed assays are naturally treated as noisy measurements of a latent positive curve. A simple starting point is multiplicative log-normal error,

$$\log y_{ij} = \log c_i(t_{ij}) + \epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0, \sigma^2_{\log}). \tag{7}$$

Lower and upper quantification limits can be represented as censoring rather than replacing censored observations by exact boundary values.

Between-participant variation can be introduced through log-normal random effects on peak level and decay rates, with possible correlation between them. This permits both individual and population targets and asks whether additional participants compensate for sparse information within each trajectory. Population mechanistic models have used this type of structure in vaccine applications (Pasin et al. 2019).

Table 1 gives tentative values around which to begin discussion. They are calibration aids, not universal biological constants. In particular, assay units and protection thresholds are not automatically portable between studies.

Table 1: Tentative starting values for a COVID-19 baseline.

Quantity	Starting value	Interpretation / possible variation
Early level c_0	5000 BAU/mL	Illustrative assay scale; consider 3000–8000
Log-error SD $\sigma_{\log}$	0.20	Explore lower and higher assay variability
Fast half-life	35 days	Early contraction; tentative range 25–60 days
Slow half-life	581 days	Highest-priority tail sensitivity; consider 250–900 days
Switching time	75 days	Tentative transition between fast and slow components
Fast:slow contribution	about 80:20	Implied by the smooth Hogan parameterisation
Protection anchor	20.2% / 3.0%	Convalescent neutralisation for 50% symptomatic / severe protection
Peak / rate log-SD	0.30 / 0.25	Moderate between-participant heterogeneity
Peak–rate correlation	−0.30	Plausible starting value; include zero as a comparison

The kinetic starting values draw principally on the model-based analysis of Hogan et al. (2023); the protection anchors come from Khoury et al. (2021). The workshop may choose a different assay normalisation or tail calibration before any headline analysis.

5 A possible analysis programme

A simulation-first programme would create a full longitudinal dataset from a known trajectory, then reveal only the portion that a candidate study would have observed. Each candidate model could be fitted to the reduced data and extrapolated to T^*. Because the generating truth is known, the exercise can separate uncertainty from confident misspecification.

A natural progression is:

1. establish a transparent homogeneous baseline in which a biphasic model is fitted to biphasic data;
2. fit all candidate families to the same truncated data to expose disagreement in the tail;
3. generate data from a richer mechanistic model and ask whether simple fitted models remain useful;
4. introduce participant heterogeneity and compare individual with population duration;
5. if useful, map antibody trajectories to symptomatic and severe-disease protection or compare the framework with an open real dataset.

Points for discussion include which duration is clinically meaningful; how strongly to privilege COVID-19; which model families and parameterisations are worth pursuing; how inference and study design should be handled; which comparisons would be most informative; and what degree of extrapolation would be acceptable to a trial designer.

6 How the literature informs the project

Plotkin and Gilbert (2012). This short framework gives us the vocabulary needed to distinguish a statistical correlate, a mechanistic correlate and a surrogate endpoint, helping us avoid interpreting a protection-relevant antibody marker as the whole immune mechanism (Plotkin and Gilbert 2012).

Khoury et al. (2021). This is the main quantitative bridge from neutralisation titre to protection: it supplies the logistic relationship and relative 50%-protection anchors that can be used to turn a fitted antibody curve into an endpoint-specific protection curve (Khoury et al. 2021).

Cromer et al. (2022). The meta-analysis extends the titre-to-protection framework across variants and boosting, reminding us that starting titre and antigenic change can alter protection duration even when the underlying decay process is unchanged (Cromer et al. 2022).

Gilbert et al. (2022). The mRNA-1273 efficacy-trial analysis provides trial-level evidence that neutralisation is a graded correlate of risk, supporting continuous marker models rather than only a binary laboratory cut-off (Gilbert et al. 2022).

Feng et al. (2021). This analysis of ChAdOx1 nCoV-19 similarly supports a continuous relationship between immune markers and symptomatic or asymptomatic infection, and cautions against treating any single threshold as universal (Feng et al. 2021).

Earle et al. (2021). The cross-vaccine synthesis supplies broader evidence that antibody levels can function as a protective correlate for COVID-19 vaccines, strengthening the case for the marker-based starting point while not establishing causality by itself (Earle et al. 2021).

Hogan et al. (2023). This model-based study provides the tentative COVID-19 biphasic calibration—including fast and slow half-lives and a switching time—and illustrates how immune kinetics can be connected to longer-term effectiveness against variants (Hogan et al. 2023).

White et al. (2015). The RTS,S malaria analysis is the closest published analogue to the project's intended chain: it uses biphasic antibody kinetics and links the resulting waning marker to duration of vaccine efficacy (M. T. White et al. 2015).

Andraud et al. (2012). The nested plasma-cell and antibody models provide a clear mechanistic template for generating trajectories on several biological time scales and for asking when a simpler decay curve is an adequate approximation (Andraud et al. 2012).

Pasin et al. (2019). The Ebola prime-boost study demonstrates how mechanistic antibody dynamics and between-participant variation can be estimated together, making it a useful model for the hierarchical phase of this project (Pasin et al. 2019).

Amanna et al. (2007). Long-term measurements across antigens show that antibody durability varies greatly and need not follow a single constant half-life, motivating the power-law and plateau alternatives and explicit attention to the tail (Amanna et al. 2007).

Selected literature

Amanna, Ian J., Nichole E. Carlson and Mark K. Slifka (2007). 'Duration of humoral immunity to common viral and vaccine antigens'. In: *New England Journal of Medicine* 357.19, pp. 1903–1915. DOI: `10.1056/NEJMoa066092`.

Andraud, Mathieu, Olivier Lejeune, Jammbe Z. Musoro, Benson Ogunjimi, Philippe Beutels and Niel Hens (2012). 'Living on three time scales: the dynamics of plasma cell and antibody populations illustrated for hepatitis A virus'. In: *PLoS Computational Biology* 8.3, e1002418. DOI: `10.1371/journal.pcbi.1002418`.

Cromer, Deborah et al. (2022). 'Neutralising antibody titres as predictors of protection against SARS-CoV-2 variants and the impact of boosting: a meta-analysis'. In: *The Lancet Microbe* 3.1, e52–e61. DOI: `10.1016/S2666-5247(21)00267-6`.

Earle, Kristen A. et al. (2021). 'Evidence for antibody as a protective correlate for COVID-19 vaccines'. In: *Vaccine* 39.32, pp. 4423–4428. DOI: `10.1016/j.vaccine.2021.05.063`.

Feng, Shuo, Daniel J. Phillips, Thomas White et al. (2021). 'Correlates of protection against symptomatic and asymptomatic SARS-CoV-2 infection'. In: *Nature Medicine* 27.11, pp. 2032–2040. DOI: `10.1038/s41591-021-01540-1`.

Gilbert, Peter B., David C. Montefiori, Adrian B. McDermott et al. (2022). 'Immune correlates analysis of the mRNA-1273 COVID-19 vaccine efficacy clinical trial'. In: *Science* 375.6576, pp. 43–50. DOI: `10.1126/science.abm3425`.

Hogan, Alexandra B. et al. (2023). 'Estimating long-term vaccine effectiveness against SARS-CoV-2 variants: a model-based approach'. In: *Nature Communications* 14.1, 4325. DOI: `10.1038/s41467-023-39736-3`.

Khoury, David S. et al. (2021). 'Neutralizing antibody levels are highly predictive of immune protection from symptomatic SARS-CoV-2 infection'. In: *Nature Medicine* 27.7, pp. 1205–1211. DOI: `10.1038/s41591-021-01377-8`.

Pasin, Chloé et al. (2019). 'Dynamics of the humoral immune response to a prime-boost Ebola vaccine: quantification and sources of variation'. In: *Journal of Virology* 93.18, e00579–19. DOI: `10.1128/JVI.00579-19`.

Plotkin, Stanley A. and Peter B. Gilbert (2012). 'Nomenclature for immune correlates of protection after vaccination'. In: *Clinical Infectious Diseases* 54.11, pp. 1615–1617. DOI: 10.1093/cid/cis238.

White, Michael T., Robert Verity, Jamie T. Griffin et al. (2015). 'Immunogenicity of the RTS,S/AS01 malaria vaccine and implications for duration of vaccine efficacy: secondary analysis of data from a phase 3 randomised controlled trial'. In: *The Lancet Infectious Diseases* 15.12, pp. 1450–1458. DOI: 10.1016/S1473-3099(15)00239-X.